form of entropy, because the number of arrangements in such a situation is given by the product of the individual arrangements, so the logarithm of the number of arrangements is additive.

7. While we can, *in principle*, predict where each page will land, you might be concerned that there is an additional element that determines the page ordering: how you gather the pages together in a neat stack. This is not relevant to the physics being discussed, but in case it bothers you, imagine that we agree that you'll pick up the pages, one by one, starting with the one that's closest to you, and then picking up the page closest to that one, and so on. (And, for example, we can agree to measure distances from the nearest corner of the page in question.)

8. To succeed in calculating the motion of even a few pages with the accuracy required to predict their page ordering (after employing some algorithm for stacking them in a pile, such as in the previous note), is actually *extremely* optimistic. Depending on the flexibility and weight of the paper, such a comparatively "simple" calculation could still be beyond today's computational power.

9. You might worry that there is a fundamental difference between defining a notion of entropy for page orderings and defining one for a collection of molecules. After all, page orderings are discrete—you can count them, one by one, and so although the total number of possibilities might be large, it's finite. To the contrary, the motion and position of even a single molecule are continuous—you can't count them one by one, and so there is (at least according to classical physics) an infinite number of possibilities. So how can a precise counting of molecular rearrangements be carried out? Well, the short response is that this is a good question, but one that has been answered fully—so if that's enough to ease your worry, feel free to skip what follows. The longer response requires a bit of mathematics, so without background this may be tough to follow completely. Physicists describe a classical, many-particle system, by invoking *phase space*, a 6N-dimensional space (where N is the number of particles) in which each point denotes all particle positions and velocities (each such position requires three numbers, as does each velocity, accounting for the 6N dimensionality of phase space). The essential point is that phase space can be carved up into regions such that all points in a given region correspond to arrangements of the speeds and